Wildfire Ops

github.com/jucamohedano/wildfire-response-env

Project report · 29 October 2026 · source revision 6e9818e

Run a wildfire control centre for 24 simulated hours

One coordinating agent receives incomplete information, assigns finite resources and protects people. Its actions change a seeded world; recorded outcomes produce a reproducible score.

1 HANDOVER

Take over an established fire

Start with an aged fire, deployed resources, existing commitments and a timestamped handover.

1,440 model minutes · one-minute world ticks

2 PUBLIC EVIDENCE

Work with what has arrived

Read resident and crew reports, weather, resource telemetry, destination status and acknowledgments.

Observation time and receipt time stay separate

3 TOOLS & ORDERS

Choose a response

```
request_status({
  "center_id": "center-1",
  "subject": "resources"
})
```

Contract example · eight structured tools

4 CONSEQUENCES

Advance, observe and verify

Crews travel and work. Fires and population conditions evolve. The agent receives new information.

✓ Exact replay + independent reward accounting

THE PROJECT

Control-centre coordination

INCIDENT-DAY v2

Working CPU environment | Source-grounded synthetic exercises

Coordinate a changing incident with delayed information and finite resources.

The agent manages fire trucks, ground crews and fixed-wing aircraft, exchanges information with response centres, and chooses shelter or evacuation orders for population cohorts.

It selects missions, safe approaches and destinations; the router resolves travel. Mobilization, travel, setup, work, return and replenishment remain separate simulated stages.

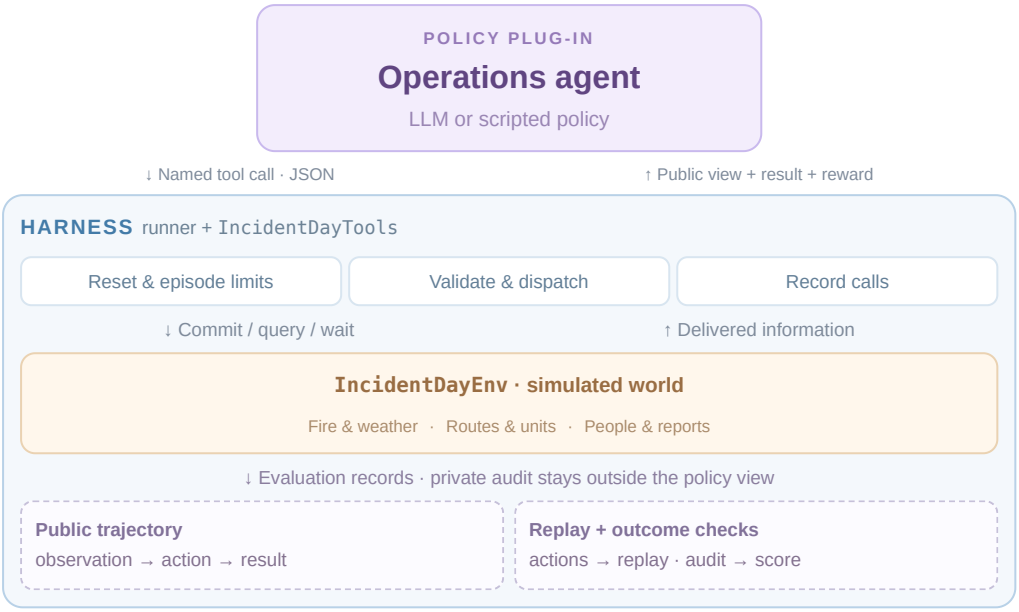
Current scope: related development exercises on one source landscape. A representative day-mode training corpus and field calibration are still needed. There is no demonstrated trained-policy improvement yet.

World horizon	Execution	Interface
24 model hours	**CPU · no API key**	**Python + HTTP/OpenEnv**

Realism focuses on event order, plausible timing, access, finite resources and consequences. The simulator's numerical behavior still needs operational validation.

The agent’s execution harness

Custom Python episode runner + named-tool layer around the simulator.



Same programmatic contract through Python or optional HTTP/OpenEnv transport.

Read delivered knowledge

`get_situation · get_reports`

Inspect approaches

`route_options · evacuation_options`

Check and act

`check_action · commit_orders`

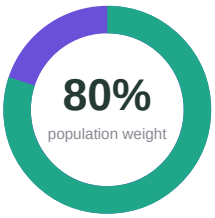
Coordinate and wait

`request_status · advance_time`

Read / check **0 min** | Commit / status **+1 min** | Wait **≤60 min** · early wake

The reward function

80% population protection · 20% terrain preserved · independently checked.



People protected

Fraction of the initially unexposed population that remains unexposed.

Terrain preserved

Fraction of initially unburned, burnable cells that avoids new burning.

$$0.8 \times P + 0.2 \times B$$

Weights, not an observed safety rate. Inherited harm and burned area are excluded. The score is computed at day end; censored or abandoned runs cannot earn a successful day score.

✓ Report active fire, burned area and unresolved evacuations alongside the score.

A high score can coexist with active fire. Reward calibration remains open in issue #27. Simulated exposure is an exercise measure, not a mortality estimate.

350 synthetic people — Two fixed cohorts



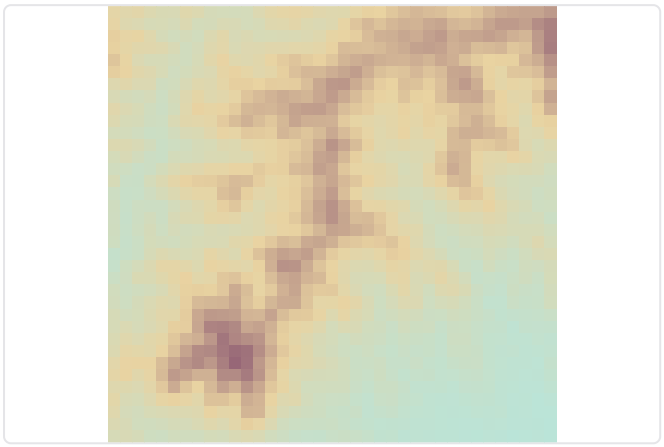
Cohort-level movement and exposure; sizes are exercise assumptions.

Real setting: maps of numbers

Sierra Bermeja, Málaga · historical context: September 2021.

Terrain

IGN · LiDAR-derived MDT05 (5 m) ↗

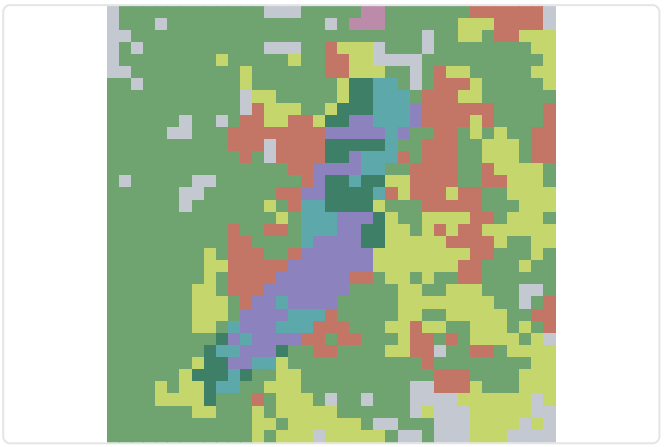


11-1,371 m · prepared mean height

ASC / GeoTIFF → 25 m samples

Vegetation / fuel

EFFIS · 250 m source pixels ↗



Model classes 1-10 · gray = source gap

Classification GeoTIFF

↓ 37 × 36 grid · 500 m cells → JSON

MA04 Estepona · IFAPA/RIA via SiAR

384 half-hour readings · 8-15 September 2021

Temperature | Humidity | Wind speed / direction | Rain

Numeric layers, separate from display imagery. Station observations support context; runtime wind remains synthetic. Weather record ↗

Prepared data: private Spain release v0.3.0 ↗ · 12 development tasks · 48 validation rows. Download guide ↗

Source fuel-class coverage

1,332 cells

1,252 source-classified

80 estimated

This measures input coverage, not physical validity. The cached land-cover query left 27 centres unclassified. Aerial review now suggests land at all 27; full-cell masks and fuel remain unverified, so runtime guards stay. Unknown fuel is never treated as a firebreak.

How synthetic data is generated

Pin the landscape, construct a scenario, run its consequences and retain a verifiable trajectory.

SOURCE SETTING

Real geography

Terrain + fuel + roads
Weather and incident context

DECLARED ASSUMPTIONS

Exercise parameters

Estimated fuel + roster + stock
Setup, refill and delay profiles

DESIGNED CASE

Choose the challenge

Fire footprint + timed changes
Access, aircraft, destinations, shortages

↓

Compile + pin identities

Align layers, keep uncertainty masks, validate supported starts and access.

↓

Warm up 60 minutes

Aged fire >500 ha + finite roster + partial handover. Synthetic prehistory, not expert actions.

↓

ONE SEEDED 24-HOUR EPISODE

Agent
Delivered knowledge
Named tools + orders

⇄

Simulated world
Travel + work + fire
People + event-triggered reports

Consequences produce observations; separate seeded reaction and delivery delays produce new messages. Queries do not resample fire or weather.

↓

Validate the complete interaction

Clocks + constraints + exact replay + independent reward + checkpoint recovery

Verified synthetic records

Public actor trajectory · private evaluator audit
Share only aggregate validation/provenance

Incomplete or failed

Retain reason; resume or fix/version
Never count an absent check as a pass

Current evidence: 12 configurations × 4 scripted policies = 48 completed runs. The Hub holds 12 task rows + 48 validation rows, not model demonstrations.

Estimate
missing map classes

Assume
operational parameters

Generate
events + communications

Keep unknown
unsupported land / access

Rules for the current incident-day exercises

1 · Pin the setting

Matching input, grid, road and scenario identities. Preserve source-known masks; label every estimate and assumption.

2 · Validate the handover

Start on supported, source-known fuel. Warm-up must remain inside the supported domain and exceed 500 ha.

3 · Respect access

Use directed mapped roads and supported walking terrain. Choose useful staging and recovery; missing access is not a shortcut.

4 · Keep resources finite

Availability, water, stock, duty and compatible facilities constrain work. Aircraft require modeled windows and clearance, not a universal minimum fire size.

5 · Order events causally

Environment changes → commands → movement/work → fire/people → delivered feedback → audit. Mobilize, arrive and set up before working.

6 · Delay information honestly

Observe → react/report → receive. Delays use separate seeded source/channel profiles and remain within the exercise's 60-minute bound.

7 · Protect actual cohorts

Conserve people; check destination capacity and access. Closure can require relocation; arrival does not confer permanent immunity.

8 · Keep knowledge separate

The policy sees delivered evidence and last-known status. Hidden fire, future schedules and evaluator records stay private; a source role is not proof of truth.

9 · Test the intended problem

A named wind/access/closure case must exercise its constraint. Gate every active configuration with replay, outcome and runtime checks; retain failed or unobserved status.

Compilation and evaluation checks: day compiler, source-map checks, event bus and runtime gate. These are exercise rules, not calibrated wildfire doctrine.

Synthetic inputs: estimates and variation

Gaps: estimates, assumptions and unknowns

Context at the 80 missing-fuel cell centres



- Woodland **24**
- Shrub **16**
- Crops **7**
- Riparian **3**
- Urban **3**
- Cached gap **27**

Fuel classes 80 cells

Estimated

Up to eight compatible neighbours vote by inverse distance. Categories are not averaged. Urban cells use a separate built/garden assumption (fuel factor 0.35).

Map context 27 centres

Visually reviewed

IGN aerial imagery suggests land at all 27 centres. Some 500 m cells contain buildings or small water features. This is a visual estimate, not historical fuel or a precise land mask; runtime guards remain. Review and source ↗

Fire-front weather

Model assumptions

Coastal observations support context. Runtime wind uses synthetic model units. Fire-front wind conversion and incident-specific moisture calibration remain unresolved.

Dispatch and response

Synthetic

Missing incident-matched logs are replaced by declared exercise inputs: roster, stocks, delays and setup/refill times. Duty windows are staffing periods; replenishment requires a compatible facility.

Terrain and access

Restricted support

Supported height crop and masks are retained. Routes use mapped corridors and declared walking assumptions. Missing links are not invented; terrain metadata and field access need evidence.

The 80 fuel estimates have low confidence. Neighbour agreement is not a probability of correctness or proof of accuracy on the missing cells. Estimation method and provenance ↗

Which variables change?

Six pinned cases per input suite. Designed combinations; not every factor is resampled.

Fire + time

Footprint · age · event schedule

Source-known starting fuel; warm-up and boundary checks

Wind + disruptions

Wind changes · lost access · aircraft windows

Destination closures · centre shortages

Reports + delays

Event templates · reaction + delivery ≤60 min

Seeded by source, role, urgency, channel + connectivity

Resources + destinations

Readiness · water · shifts · occupancy

Finite roster and compatible facilities

↓

Seeded simulation

Actions → movements → reports → outcomes

Fixed Landscape · 24 hours · 80/20 reward · cohorts of 200 + 150

Run identity: scenario version + pinned seed + input/network hashes. Arbitrary seed overrides are rejected. These are synthetic exercises, not historical transcripts.

▼ **Archived two-hour sampler: different rules and scope**

The two-hour expansion sampled ignition offsets/times, five wind vectors, report delays, duplicate counts, resource holds, request delays, readback mismatch, broadcast delay/failure and wind-shift timing. It produced 960 configuration groups with two seeded/wording variants each.

Its design seed was `20261026`; per-case seeds were `82000000 + 2 × group_index + variant`. Its additional rules checked false-alarm/no-ignition consistency, distinct multiple-fire sites, conflicting locations, duplicate lineage and configuration-group split integrity. That sampler is archived: it had night-only horizons and retained intent warnings. These factors are documented history, not a released current day-mode training corpus. Original generator and source context ↗

For a future training corpus, create separately versioned configuration groups before policy rollouts, keep related layouts/variants in one split, validate access and event constraints, then run references and replay checks. Randomization produces counterfactual exercises; it does not create new historical incidents, expert decisions or demonstrated real-world effectiveness. Next corpus work: #18 ↗

Every active configuration, every scripted policy

Recorded operation-wide CPU acceptance. Twelve related development configurations; no model-quality or field-calibration claim.

48 / 48	261	69,120	0
Full-day replay + reward checks	Exact checkpoint continuation checks	One-minute operational ticks observed	Blocking runtime contradictions

Configuration × policy

✓ = 1,440 minutes, replay, independent reward and applicable runtime checks. Missing or blocked runs never count as passes.

	Passive	Dispatcher	Conservative	Stress
Map · 000	✓	✓	✓	✓
Map · 001	✓	✓	✓	✓
Map · 002	✓	✓	✓	✓
Map · 003	✓	✓	✓	✓
Map · 004	✓	✓	✓	✓
Map · 005	✓	✓	✓	✓
Fixture · 002	✓	✓	✓	✓
Fixture · 004	✓	✓	✓	✓
Fixture · 000	✓	✓	✓	✓
Fixture · 001	✓	✓	✓	✓
Fixture · 003	✓	✓	✓	✓
Fixture · 005	✓	✓	✓	✓

Which operations did each policy exercise?

Number of configurations with an observed call or commit. Zero means unexercised by that policy; counts do not earn reward or prove successful effects.

	Passive	Dispatcher	Conservative	Stress
Read situation	12/12	12/12	12/12	12/12
Read reports	12/12	12/12	12/12	12/12
Inspect approaches	0/12	12/12	0/12	12/12
Inspect destinations	0/12	8/12	0/12	8/12
Check a draft	0/12	0/12	12/12	12/12
Request resources	0/12	12/12	0/12	12/12
Authorize allocation	0/12	12/12	0/12	12/12
Release resources	0/12	0/12	6/12	12/12
Shelter	0/12	0/12	12/12	0/12
Evacuate / relocate	0/12	8/12	0/12	8/12
Ask another centre	0/12	12/12	12/12	12/12
Advance time	12/12	12/12	12/12	12/12

Observed duration ranges

Simulated minutes. Reads and draft checks take zero. Blocked/interrupted and inherited resource spans are excluded from completed-phase ranges.

1 min	1 min	1-60 min
Commit orders	Status request	Wait / early wake
3753 completed observations	588 completed observations	10469 completed observations
1 min	**2-5 min**	**3-6 min**
Mobilization	Setup	Refill / refuel
2029 completed observations	1879 completed observations	576 completed observations
1-128 min	**3-69 min**	**3-14 min**
Outbound movement	Return movement	Status reply
2047 completed observations	825 completed observations	588 completed observations

The observer checks actual timed paths, directed road geometry, modeled aircraft clearance/windows and destination capacity. Other checks cover deliveries, finite stock, setup, replenishment, population accounting and action clocks. These rules remain exercise assumptions; they do not establish real-world access, aviation authorization or calibrated fire behavior.

One CPU worker · 1550.3 seconds · 207.6 MiB process high-water RSS. Scripted policies, not expert demonstrations. All values, pins and limitations ↗

A recorded sortie: movement takes time

First `plane-0` cycle in development case `…v2-003`.

Recorded phase intervals · 41 model minutes



Mobilize **1m** · Outbound **16m** · Setup **2m** · Return **16m** · Refill / refuel **6m**

124 min	**Mobilization begins** Recorded accepted dispatch follows authorization.
125 min	**Outbound travel starts** Modeled daylight, wind and availability gates apply.
141 min	**Aircraft reaches the sector** Arrival precedes setup and applied work.
143 min	**Drop recorded; return begins** Water and work stocks are consumed.
159 min	**Back at base; refill starts** Return travel and replenishment remain separate stages.
165 min	**Aircraft available again** Recorded refill/refuel completes the cycle.

Earlier minute-audit excerpt. Authorization/dispatch corroborate this sequence; the current runtime observer also checks modeled clearance at departures and drops. Neither establishes real aviation authorization.

Recording works; training is next

Implemented: record + verify. Planned: curate a day-mode dataset + train and evaluate a policy.

WORKING

1. Record

Public observations, tool actions, times and rewards.

WORKING

2. Verify

Replay exactly and recompute outcomes independently.

NOT IMPLEMENTED

3. Curate

New day-mode groups, untouched hold-outs and train-only cases.

NOT DEMONSTRATED

4. Learn

Compare retrieval and a trained policy on registered evaluations.

Actor traces	JSONL observations/actions with timing, terminal status and versioned reward components.
Private records	Minute audits and checkpoints for verification and deterministic interruption/resume.
Adapters	Current Python/HTTP/OpenEnv interface. AgentEnv table/runtime compatibility is demonstrated on the archived benchmark.
Future use	Verified training cases can support a frozen retrieval bank or policy optimization. Held-out outcomes must not update either.

Learning is a next stage. No active incident-day train/validation/test corpus is released yet. Historical learning recipes require explicit archival opt-in; their old cases are not silently promoted into day-mode training.

Calibration and the next evidence loop

Evidence quality is the current bottleneck; these are planned checks, not calibration results.

Calibration loop

Next evidence work

Measure	Compare	Adjust	Validate
real observations	simulation residuals	plausible ranges	different evidence

Historical perimeter + time t_0 → Reference simulation to t_1 → Observed map at t_1

Where spatial pattern **When** timing **How much** affected area

First align a historical starting perimeter and time. Keep weather uncertainty and unknown suppression explicit. Matching area alone is insufficient. Forest Service methods ↗ · NIST timeline example ↗

Current boundaries

Verified — Exact replay, independent reward, modeled constraints and installed-runtime checks.

Pending — Incident-matched calibration, terrain/source gaps, a grouped training corpus and evidence of learning improvement.

Archived — Four catalogs and 20 superseded day cases remain available for explicit replay. Current cases are development probes.

Replay establishes consistency; calibration needs independent evidence. Historical dispatcher actions are not assumed optimal. Public-release rights remain unresolved.

Delivered: operation-wide runtime validation

1 · Cover all cases

2 · Bound routes + phases

3 · Check invariants + faults

4 · Replay + resume across policies

The CPU gate is complete: 48/48 days pass. Next: separately grouped training/held-out cases and further calibration evidence; model learning is not yet demonstrated. Implementation plan ↗

▼ Reproducibility snapshot and source boundaries

Project snapshot `62eae5a7f6206bede7d78fd781155c7b404dc2a5`. All numerical acceptance evidence refers to the pinned implementation below. Documentation changes do not rescore an episode.

`Implementation: af4adc0b2fb92282246411dbac83c1eeac23eb3b4730200c0b34dc288b59323a`

`Prepared day inputs: 50031b3029dcee0a1367ccdd8125d081d19d3b500b2dd373e30ef127d47a2c4d`

`Road network: 12782cb4ccb4bbfd2d59eaaa7380ad6b1a807320cf5a15b6d01e44d7ec89fc16`

`Operation validator: 6a731cf8784eb42d9c475e3ddf8a0312d662c5ba1cb16b189b5179e197f37b8e`

The Spanish incident anchor is 8 September 2021; source layers have their own editions. Generated reports, operational choices and counterfactual outcomes are synthetic. This report contains aggregate evidence and a development-cycle excerpt; private audit files are not embedded.

Repository	Agent contract	Verification evidence	Historical anchor
Implementation and version history	API, timing, storage and reproduction	Measured checks and exact limitations	Copernicus · Sierra Bermeja, 2021
`jucamohedano/wildfire-response-env` ↗	`docs/incident-day.md` ↗	`runtime-validation-20261029` ↗	`EMSR545 activation` ↗